

DUBBO, Australia

WMO: 957190

Lat: 32.2206S

Lon: 148.5753E

Elev: 935

StdP: 14.21

Time Zone: 10.00 (AUS)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	30.6	33.2	20.8	16.1	69.3	24.5	19.2	62.8	22.5	56.7	20.4	56.8	4.7	140	0.531

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.0	100.6	67.3	97.0	66.8	93.9	66.4	72.7	87.0	71.4	85.4	70.1	84.4	13.0	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.7	112.9	75.3	67.8	105.9	74.9	66.1	99.7	74.6	37.0	86.3	35.8	85.8	34.7	84.4	80.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 23.1	20.7	18.6	DB	26.0	106.6	1.6	3.7	24.9	109.3	23.9	111.4	23.0	113.5	21.8	116.2	(4)
(5)			WB	25.6	75.5	1.8	2.0	24.3	76.9	23.2	78.1	22.2	79.2	20.8	80.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.3	78.6	76.8	71.7	64.1	55.6	50.5	48.5	50.3	56.6	63.2	69.8	74.7		
	DBStd	11.94	6.37	5.91	5.69	5.77	5.29	4.71	4.34	4.62	5.64	6.62	7.16	6.15		
	HDD50	207	0	0	0	1	13	50	81	54	7	1	0	0		
	HDD65	2200	1	1	12	81	293	435	512	456	258	115	31	5		
	CDD50	5060	886	752	673	425	186	66	35	62	205	410	595	765		
	CDD65	1580	422	333	220	55	2	0	0	0	7	60	176	305		
	CDH74	18842	5192	3645	2174	613	25	0	0	6	219	945	2236	3787		
	CDH80	9141	2900	1861	891	136	1	0	0	0	44	329	1045	1934		
(14) Wind	WSAvg	9.6	11.3	11.0	10.2	9.0	8.1	8.4	8.1	8.5	9.2	10.0	10.7	10.8		
(15) Precipitation	PrecAvg	24.3	2.8	2.0	1.9	1.6	2.1	1.5	1.9	2.0	1.9	2.2	2.0	2.1		
	PrecMax	46.1	8.7	10.6	7.4	11.5	8.7	4.2	5.4	5.7	6.3	8.3	5.8	4.5		
	PrecMin	13.0	0.0	0.2	0.0	0.0	0.1	0.0	0.1	0.0	0.0	0.1	0.0	0.0		
	PrecStd	7.3	2.2	2.0	1.7	2.2	1.7	1.0	1.3	1.2	1.5	1.6	1.5	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	105.8	105.1	96.4	88.0	76.9	69.4	68.1	73.8	84.6	91.8	102.2	102.2		
		MCWB	68.9	70.4	66.6	63.6	60.1	56.4	53.4	55.5	58.4	61.9	68.1	65.6		
	2%	DB	101.1	98.8	91.9	83.9	73.6	66.1	64.7	69.7	79.9	87.6	95.3	97.2		
		MCWB	67.9	68.4	65.6	62.1	58.4	55.2	52.6	54.2	57.8	61.0	64.1	65.5		
	5%	DB	97.4	94.6	88.5	80.6	71.0	63.5	62.3	66.2	75.6	83.9	90.3	93.9		
		MCWB	67.6	67.7	65.3	60.9	56.8	53.7	52.0	52.9	56.8	60.5	63.4	65.2		
	10%	DB	93.7	90.3	85.0	77.3	67.9	60.8	59.4	62.9	71.5	79.4	86.0	90.3		
		MCWB	67.6	67.0	64.7	60.0	55.6	52.4	50.8	51.6	55.9	59.7	62.9	64.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.0	75.2	73.1	67.2	63.6	59.4	59.6	59.7	64.2	67.9	71.4	72.6		
		MCDB	89.0	90.0	84.0	77.4	70.9	63.8	63.1	66.5	76.0	80.6	87.7	86.2		
	2%	WB	72.5	72.9	70.5	64.9	61.0	57.4	55.9	57.1	61.3	65.7	69.2	70.6		
		MCDB	88.5	86.4	81.1	76.6	67.9	61.9	60.3	64.4	72.4	79.0	84.3	84.5		
	5%	WB	71.1	71.4	68.6	63.3	59.0	55.7	53.6	54.8	59.4	63.6	67.0	69.1		
		MCDB	87.2	84.3	80.7	75.1	67.1	61.0	59.4	63.0	69.6	76.3	81.3	83.2		
	10%	WB	70.0	69.9	67.2	61.8	57.3	53.9	51.8	52.8	57.5	61.7	65.2	67.6		
		MCDB	85.6	83.6	79.3	73.3	65.9	59.6	57.4	61.3	67.5	74.1	79.0	82.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	26.0	24.5	24.3	24.9	23.5	20.7	22.0	24.6	26.8	27.2	26.3	26.8		
		MCDBR	30.4	29.5	28.4	27.8	26.6	23.2	25.5	29.6	33.6	33.4	32.7	31.4		
	5% WB	MCWBR	9.3	9.4	10.6	11.5	13.2	13.3	15.1	16.3	15.7	13.8	11.4	10.9		
		MCWBR	25.5	23.3	22.9	23.1	20.7	18.5	19.7	23.7	25.4	26.6	25.5	25.2		
(40) Clear-Sky Solar Irradiance	taub		0.352	0.350	0.324	0.306	0.289	0.285	0.280	0.288	0.318	0.329	0.348	0.342		
		taud	2.506	2.534	2.595	2.607	2.607	2.622	2.623	2.581	2.479	2.483	2.480	2.523		
	Ebn at Noon		313	307	305	294	284	278	285	297	302	310	312	317		
		Edh at Noon	36	34	30	26	23	22	23	27	33	35	37	36		
(44) All-Sky Solar Radiation	RadAvg		2426	2137	1804	1443	1084	843	957	1254	1694	2046	2249	2453		
	RadStd		127	157	130	96	61	63	59	84	110	151	213	128		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	-160	N/A	N/A	N/A
(47) Regional (13 neighbors)		+0.85	N/A	N/A	N/A	N/A	N/A	N/A	-158	+344	+202	N/A

Nomenclature: See separate page